

Experiment no. 3

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State space formulation and PEAS properties for AI application for case study in Experiment 2	
Learning Objective:	Give State space formulation and PEAS properties for AI application in Experiment 2
Learning Outcome:	Students will be able to give the PEAS properties and state space formulation for the case study in Experiment 2
Course Outcome:	CSL 604.3
Program Outcome:	(PO 3) Design/ development of solutions: Breadth and uniqueness of engineering problems i.e. the extent to which problems are original and to which solutions have previously been identified or codified (PO 12) Lifelong Learning
Bloom's Taxonomy Level:	Analysis, Create
PEASE Properties:	PEAS PROPERTIES PEAS is a framework used to specify the structure of an intelligent agent in AI. It breaks down the agent's interaction with the environment into four key components:

Performance Measure: The criteria that define the success of the agent's actions.

Environment: The surroundings or the context in which the agent operates.

Actuators: The mechanisms through which the agent interacts with the environment.

Sensors: The tools the agent uses to perceive its environment.

By defining these elements, PEAS provides a clear outline for designing and evaluating intelligent systems, ensuring they are equipped to perform their tasks effectively.

PEAS PROPERTIES FOR THE CASE STUDY

- **PERFORMANCE EVALUATION**

Performance Measure:-

The Performance Measure defines how the success of the AI system is evaluated. Since this is a Trustworthy AI system based on Multi-Objective Decision Making (MODeM), multiple criteria are considered instead of only accuracy.

1. Prediction Accuracy

- Measures how correctly the model classifies income as $>50K$ or $\leq 50K$.
- Calculated using accuracy, precision, recall, and F1-score.
- Ensures the model is technically reliable.

2. Fairness Metric

- Measures bias across sensitive attributes such as gender and race.
- Common metrics:
 - Demographic Parity Difference
 - Equal Opportunity Difference
- Ensures that predictions do not unfairly favor or disadvantage any group.

3. Explainability Score

- Evaluates how understandable the model decisions are.
- Methods used:
 - Feature importance ranking

- **Decision tree visualization**
- **SHAP/LIME explanations**
- **Important for transparency and accountability.**

4. Safety and Risk Minimization

- **Ensures the system does not produce harmful or highly biased outcomes.**
- **Minimizes critical misclassifications.**
- **Maintains robustness against noisy or incomplete data.**

5. Computational Efficiency

- **Measures time and memory usage.**
- **Important for real-time or large-scale deployment.**

Overall Performance Objective

The goal is to balance all these measures using multi-objective optimization rather than maximizing only one metric. A Pareto-optimal solution is selected that provides acceptable accuracy while maintaining fairness and transparency.

- **ENVIRONMENT**

The Environment refers to the external conditions and data within which the AI agent operates.

1. Data Environment

- **The UCI Adult Income Dataset**
- **Socio-economic attributes of individuals**
- **Contains historical bias and imbalanced data**

2. User Interaction Environment

- **Users input demographic and employment details**
- **System processes structured tabular data**

3. Regulatory and Ethical Environment

- **Must comply with fairness standards**
- **Should follow ethical AI guidelines**
- **Should avoid discrimination**

Environment Characteristics

1. Partially Observable

- **The system does not know the true income beyond the dataset.**
- **It only observes available attributes.**

2. Stochastic

- **Income prediction involves uncertainty.**
- **Same attributes may not always guarantee same outcome.**

3. Static (during prediction)

- **Environment does not change while making a single prediction.**

4. Episodic

- **Each prediction is independent of previous predictions.**

5. Single-Agent System

- **Only one AI model performs the prediction task.**

- **ACTUATORS**

Actuators are the components through which the AI agent acts upon the environment.

In this case study, actuators include:

1. Income Prediction Output

- Classifies input as:
 - Income $\leq 50K$
 - Income $> 50K$

2. Explanation Generator

- Displays reasoning behind prediction
- Highlights important features influencing decision

3. Model Update Mechanism

- Updates weights and parameters during training
- Applies optimization adjustments

4. Fairness Adjustment Module

- Reweights training samples
- Applies bias mitigation techniques

5. Reporting and Logging

- Stores performance metrics
- Reports fairness and accuracy statistics

These actuators allow the AI system to respond intelligently and responsibly.

- **SENSORS**

Sensors collect input data from the environment for processing.

In the Adult Income Prediction system, sensors gather structured attributes such as:

- Age
- Workclass
- Education level
- Marital status

	• Occupation • Relationship status • Race • Gender (Sensitive attribute) • Hours per week • Native country Sensor Characteristics • Data is structured and tabular. • Includes both numerical and categorical features. • Contains sensitive attributes used for fairness evaluation. The sensors enable the system to perceive socio-economic characteristics and use them for classification.
Data Set :	Adult Income Dataset The case study uses the Adult Income Dataset, a widely recognized benchmark dataset for classification and fairness analysis in Artificial Intelligence research. The objective of the dataset is to predict whether an individual's annual income exceeds \$50K per year based on demographic and employment attributes.
	Dataset Characteristics • Total Instances: Approximately 48,842 • Number of Input Attributes: 14 • Target Variable: 1 (Income) • Data Type: Structured, tabular dataset • Attribute Type: Combination of numerical and categorical features • Contains sensitive attributes (e.g., gender, race) for fairness analysis
	Dataset Columns (Attributes) The dataset used in this experiment contains the following columns: 1. age – Age of the individual (Numerical)

	2. workclass – Employment type (Private, Self-employed, Government, etc.) 3. fnlwgt – Final sampling weight assigned by census bureau (Numerical) 4. education – Highest level of education attained (Categorical) 5. educational-num – Numerical representation of education level 6. marital-status – Marital status (Married, Single, Divorced, etc.) 7. occupation – Type of occupation 8. relationship – Relationship status within household 9. race – Race of the individual (Sensitive attribute) 10. gender – Gender (Sensitive attribute) 11. capital-gain – Capital gains earned in a year (Numerical) 12. capital-loss – Capital losses incurred in a year (Numerical) 13. hours-per-week – Number of working hours per week (Numerical) 14. native-country – Country of origin (Categorical) 15. income – Target variable • $\leq 50K$ • 50K DATASET LINK
	Sensitive Attributes For the Trustworthy AI case study, the following attributes are considered sensitive: • gender • race These attributes are used to measure fairness and detect bias in predictions.
Methods:	The methodology is based on Multi-Objective Decision Making (MODeM) for designing a Trustworthy AI system.

Step 1: Problem Definition

Develop a classification model to predict income while optimizing:

- Accuracy
- Fairness
- Explainability
- Safety

Step 2: Model Selection

Machine Learning models used:

- Logistic Regression
- Decision Tree
- Random Forest

Step 3: Define Objective Functions

- Maximize Accuracy
- Minimize Fairness Difference
- Maximize Interpretability
- Maintain Low Error in sensitive cases

Step 4: Multi-Objective Optimization

- Weighted Sum Method or Pareto Optimization applied
- Trade-offs analyzed between objectives
- Generation of Pareto-optimal solutions

Step 5: Fairness Enhancement

- Bias detection using demographic parity
- Reweighting or constraint-based fairness methods

Step 6: Explainability Integration

- Feature importance analysis
- Decision tree visualization
- Model interpretation tools

	Step 7: Evaluation Performance compared using: • Accuracy • Precision, Recall, F1-score • Fairness metrics • Confusion Matrix This method ensures that the AI system is both technically strong and socially responsible.
Outcome :	The implementation produced the following results: 1. Balanced Performance • High classification accuracy achieved • Fairness gap between gender and race reduced 2. Pareto-Optimal Solutions • Multiple optimal trade-off models generated • Decision-makers can choose models based on priority 3. Improved Trustworthiness • Reduced bias in predictions • Transparent decision-making process • Better interpretability using decision trees 4. Practical Insight The study demonstrates that optimizing only accuracy can lead to biased systems, while multi-objective optimization produces socially acceptable AI solutions.
Conclusion :	The case study on Adult Income Prediction using Multi-Objective Decision Making highlights the importance of designing Trustworthy AI systems. Instead of focusing only on prediction accuracy, the system balances: • Fairness • Explainability • Safety

	• Performance The results show that multi-objective optimization provides balanced and ethically responsible AI models. This supports engineering design principles (PO 3) and emphasizes continuous learning in responsible AI development (PO 12). Trustworthy AI is essential for real-world deployment, especially in sensitive domains such as finance, healthcare, and governance.
References:	Dua, D. and Graff, C. (2019). <i>UCI Machine Learning Repository</i>. University of California, Irvine, School of Information and Computer Sciences. (Adult Data Set). • Russell, S. and Norvig, P. (2021). <i>Artificial Intelligence: A Modern Approach</i>, 4th Edition. Pearson Education. • Barocas, S., Hardt, M., & Narayanan, A. (2019). <i>Fairness and Machine Learning</i>. • IEEE (2019). <i>Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems</i>. • Mehrabi, N. et al. (2021). "A Survey on Bias and Fairness in Machine Learning." <i>ACM Computing Surveys</i>.